

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of specific leaf area (SLA) per plot (esy4 = 4qm plots)

MB

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
```

```
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
library(ggbeeswarm)
library(patchwork)
library(DHARMA)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.sla)
```

Load data

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 234.5920 230.8435 227.0950
```

```
median(sites$y)
```

```
## [1] 232.03
```

```
sd(sites$y)
```

```
## [1] 32.48951
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
## 173.985 282.840
```

```
sites %>% distinct(id.site) %>% nrow()
```

```
## [1] 119
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct> <int>
## 1 654      101
## 2 664       91
## 3 686       99
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord>      <int>
## 1 positive      45
## 2 restored     221
## 3 negative      25
```

```
sites %>% count(esy4)
```

```
## # A tibble: 2 x 2
##   esy4      n
##   <fct> <int>
## 1 R22     210
## 2 R1A      81
```

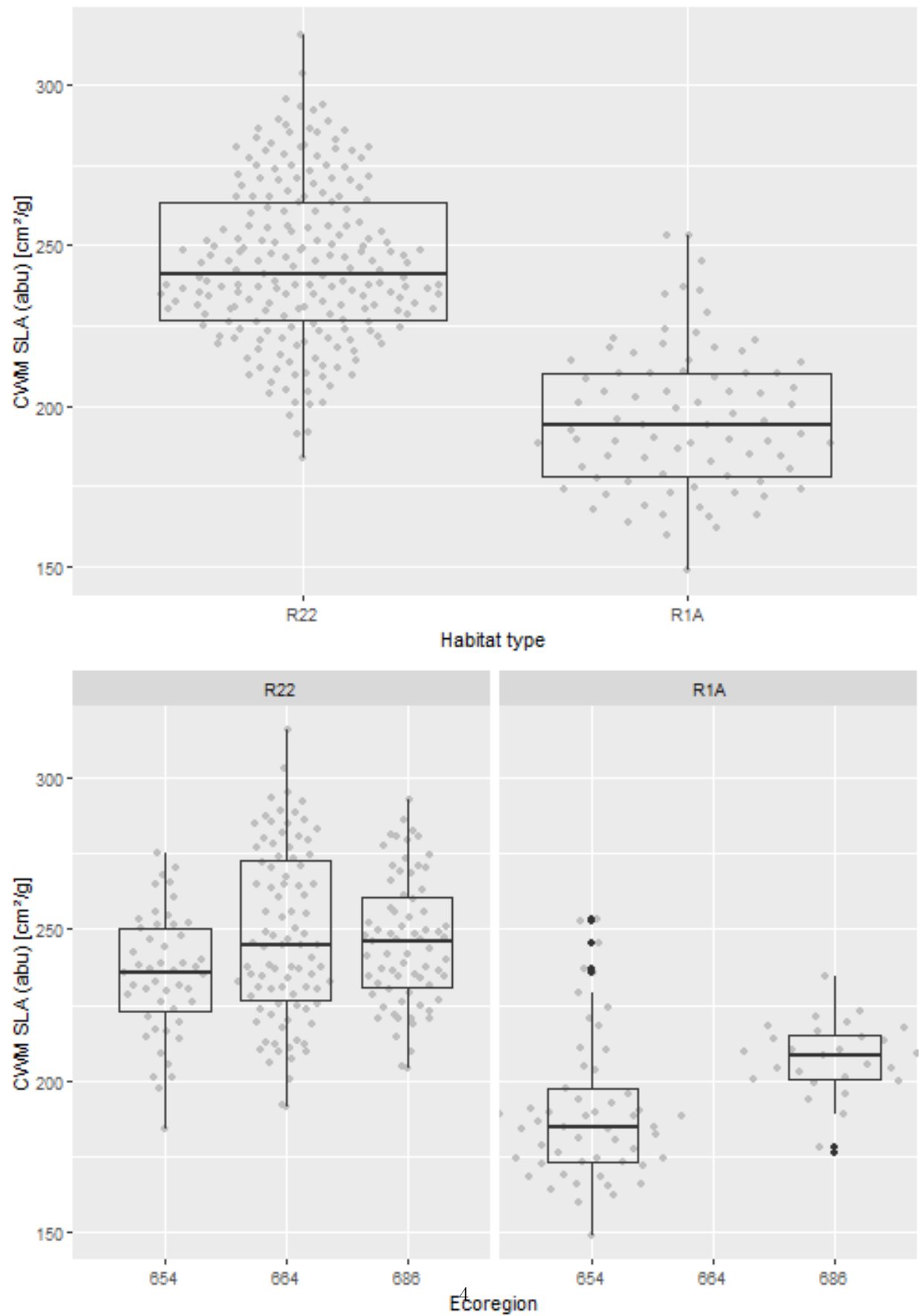
```
sites %>% count(esy4, eco.id)
```

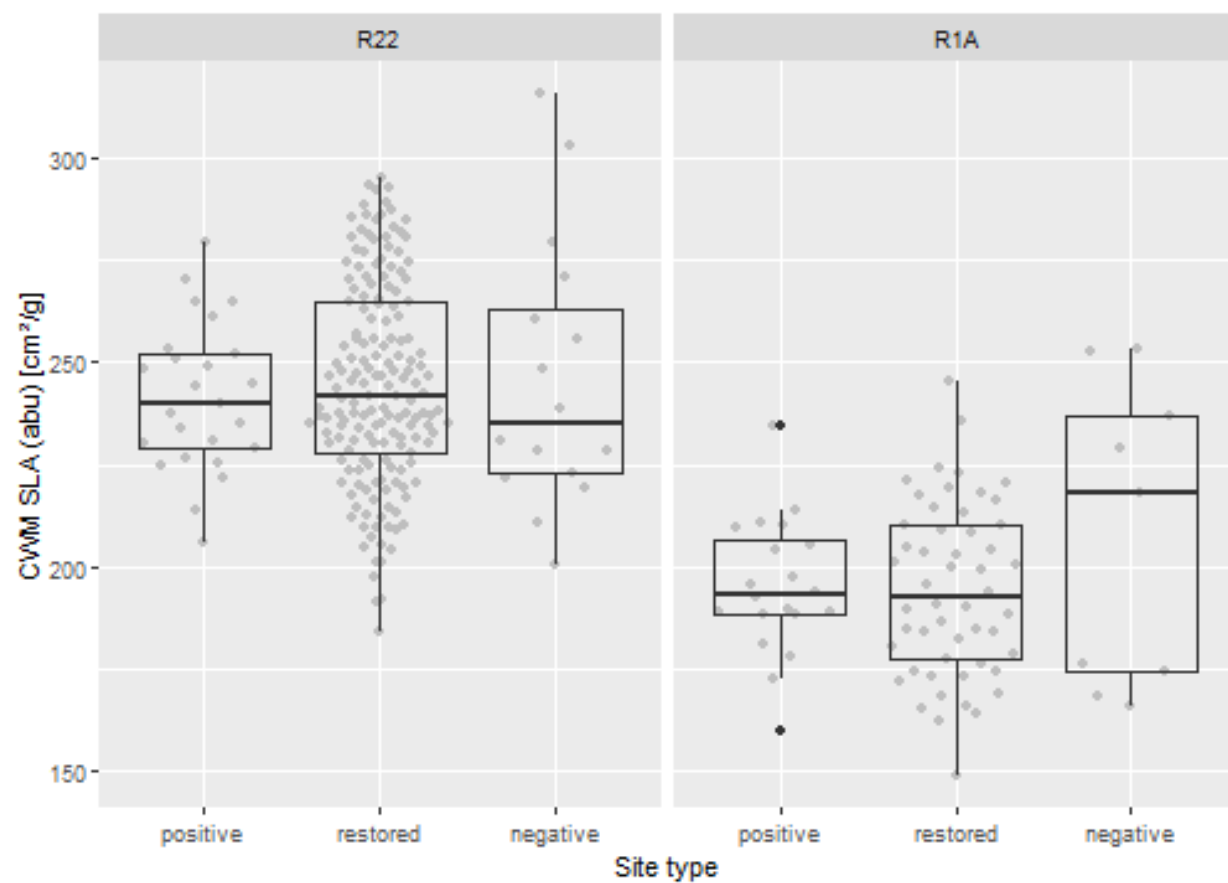
```
## # A tibble: 5 x 3
##   esy4 eco.id      n
##   <fct> <fct> <int>
## 1 R22  654      48
## 2 R22  664      91
## 3 R22  686      71
## 4 R1A  654      53
## 5 R1A  686      28
```

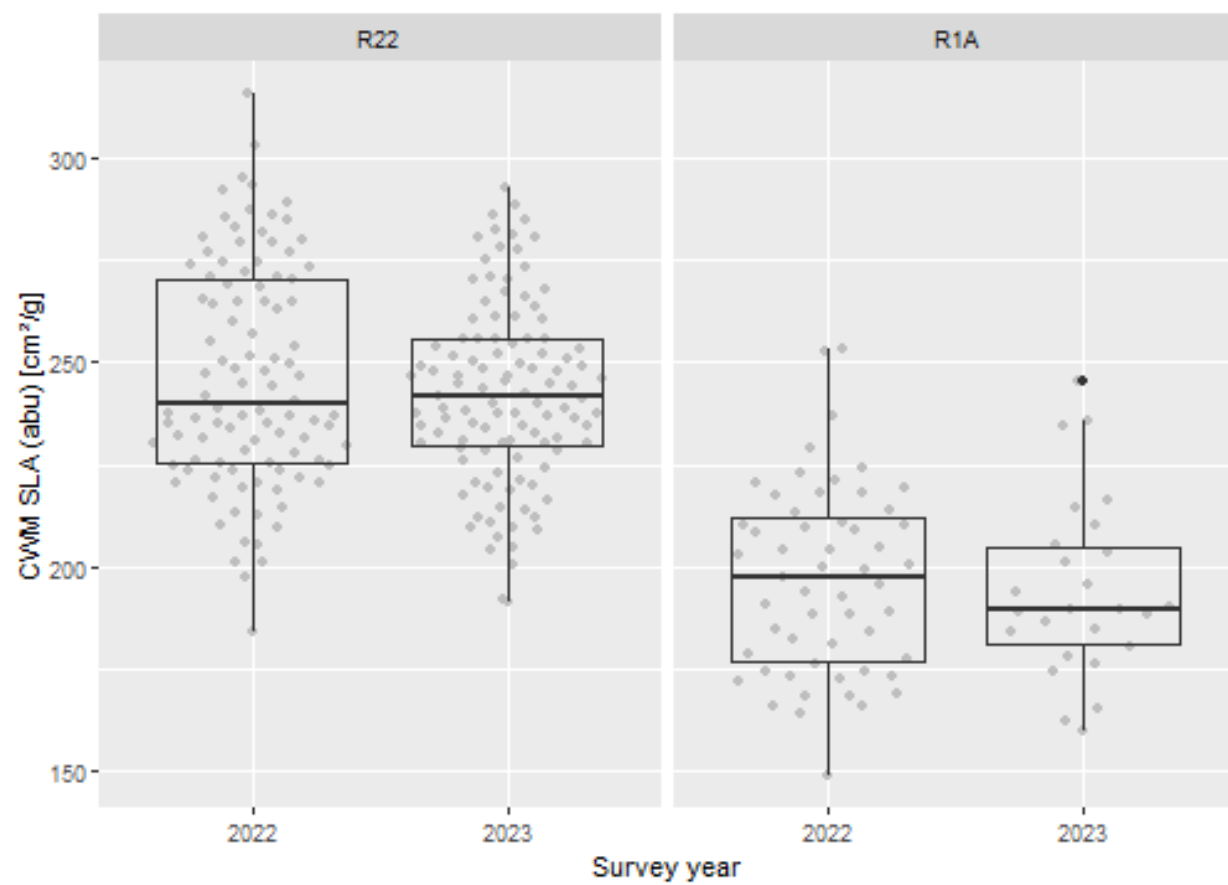
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 6 x 3
##   esy4 site.type      n
##   <fct> <ord>      <int>
## 1 R22  positive      25
## 2 R22  restored     169
## 3 R22  negative      16
## 4 R1A  positive      20
## 5 R1A  restored      52
## 6 R1A  negative       9
```

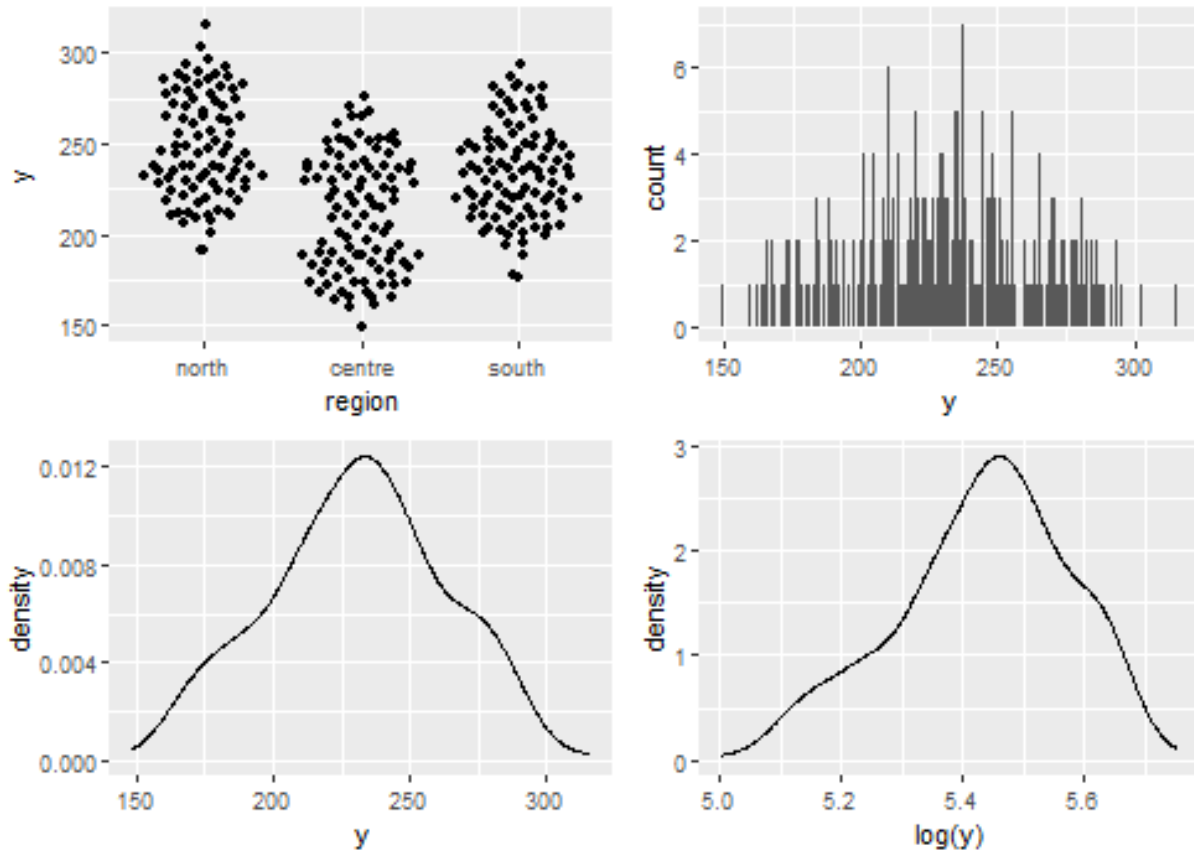
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 *Ecography* DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum.)) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_sla_cwm_2.Rdata"))
load(file = here("outputs", "models", "model_sla_cwm_3.Rdata"))
m_1 <- m2
m_2 <- m3

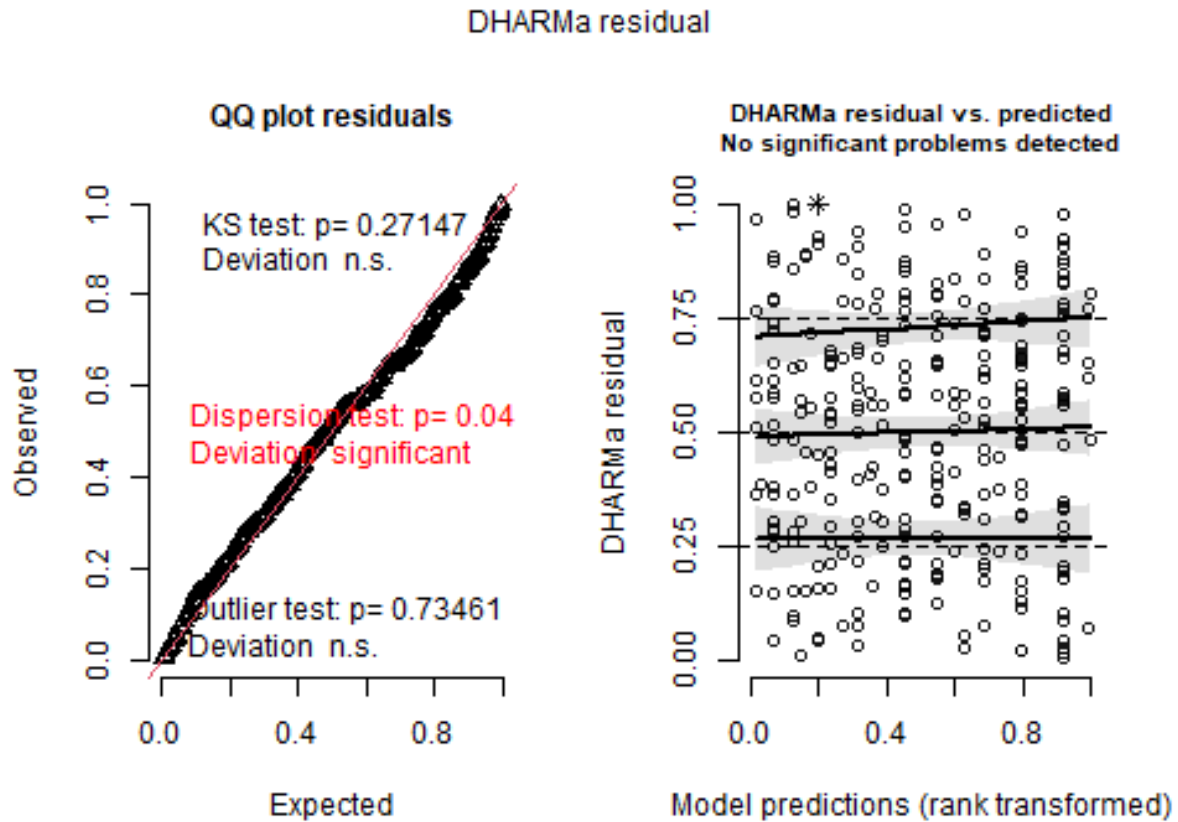
m_1@call
## lmer(formula = sqrt(y) ~ (esy4 + site.type + eco.id + obs.year)^2 +
##       (1 | id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = sqrt(y) ~ (esy4 + site.type + eco.id + obs.year +
##      hydrology)^2 + (1 | id.site), data = sites, REML = FALSE)
```

Model check

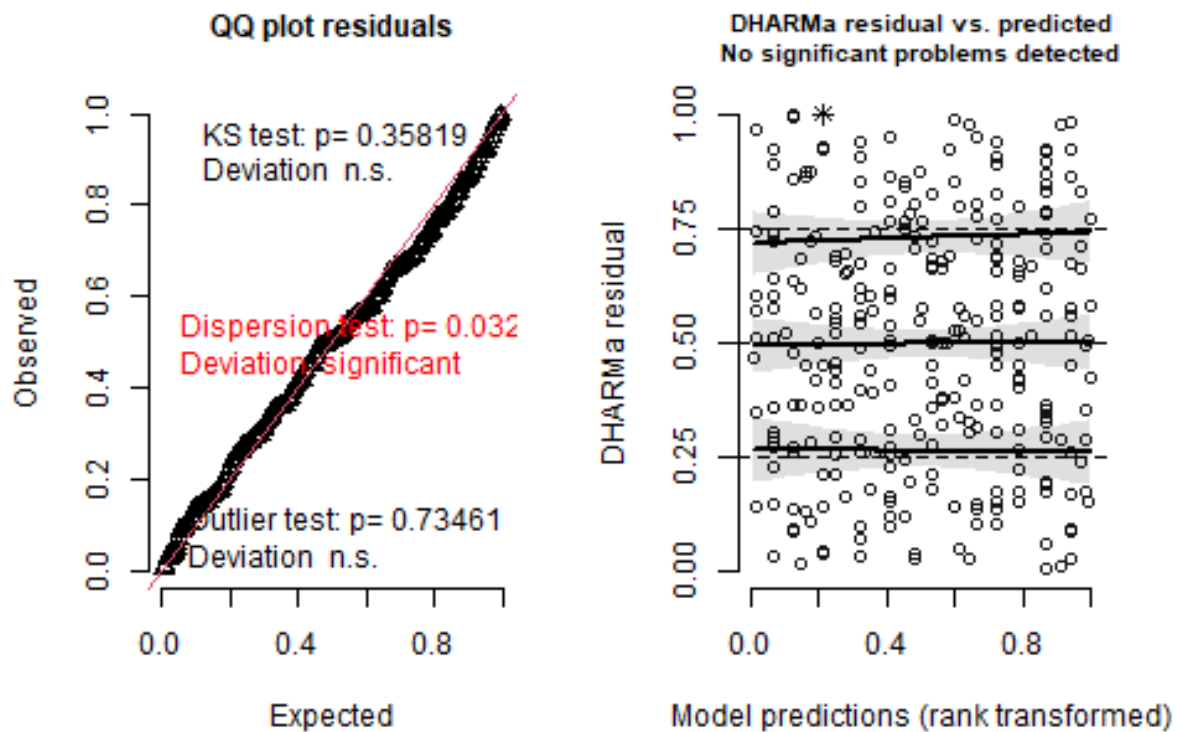
DHARMA

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

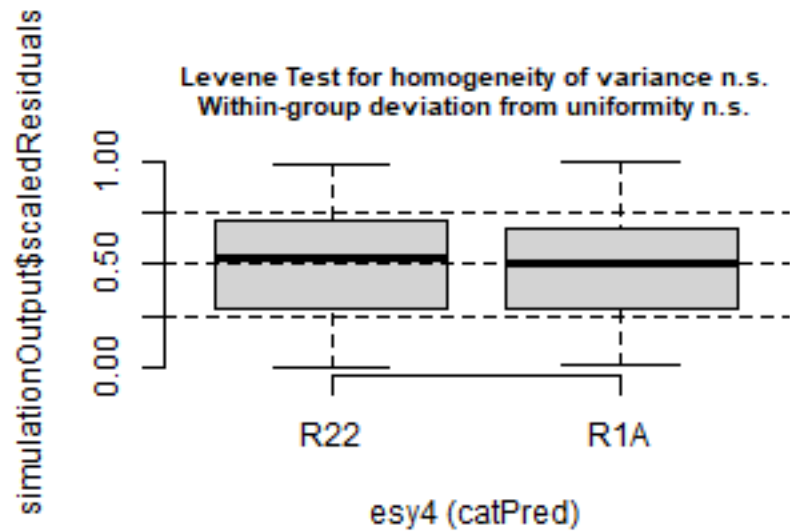


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

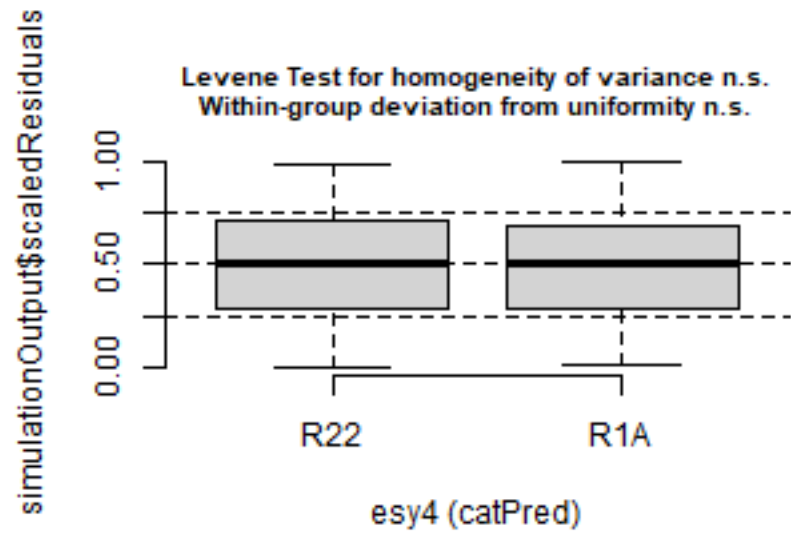

DHARMa residual



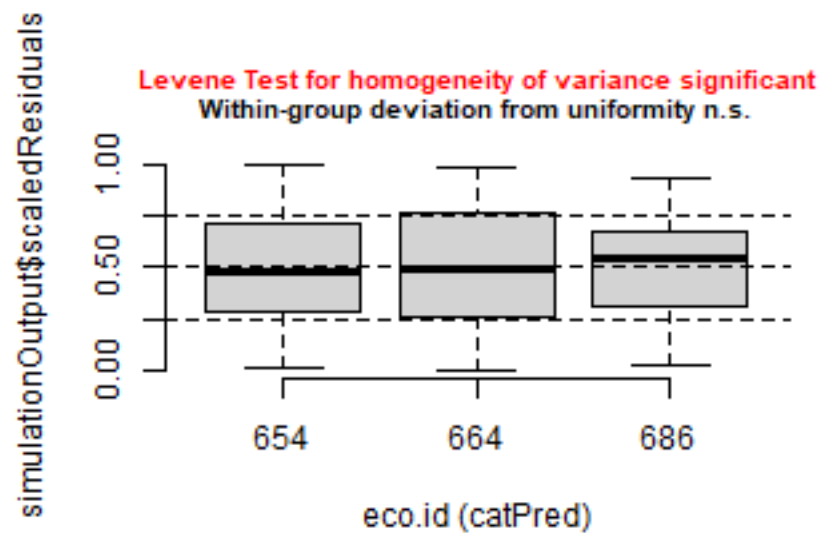
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



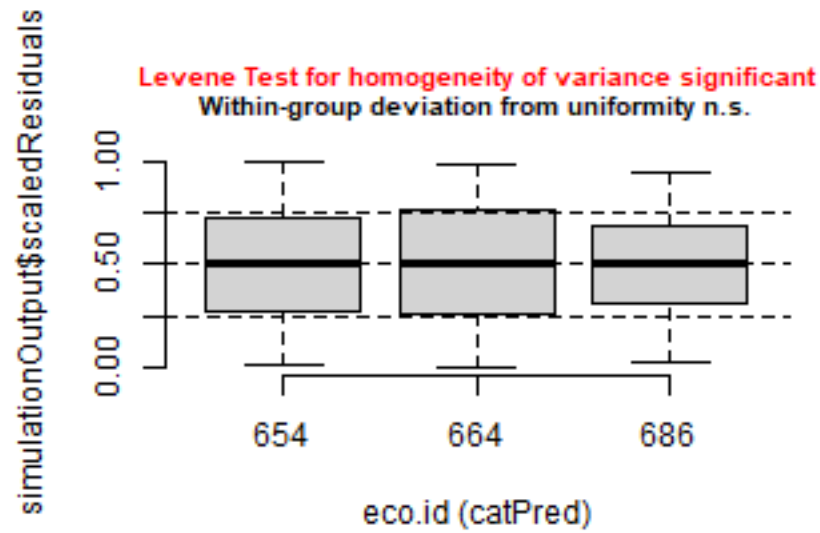
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



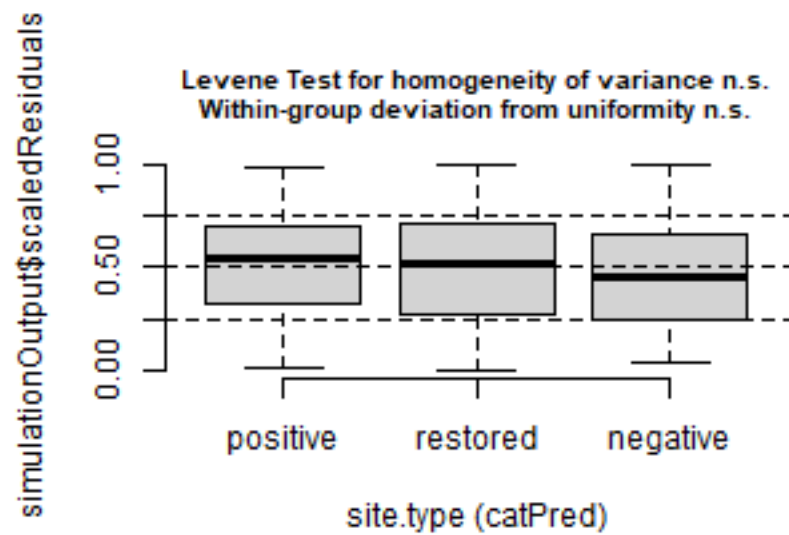
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



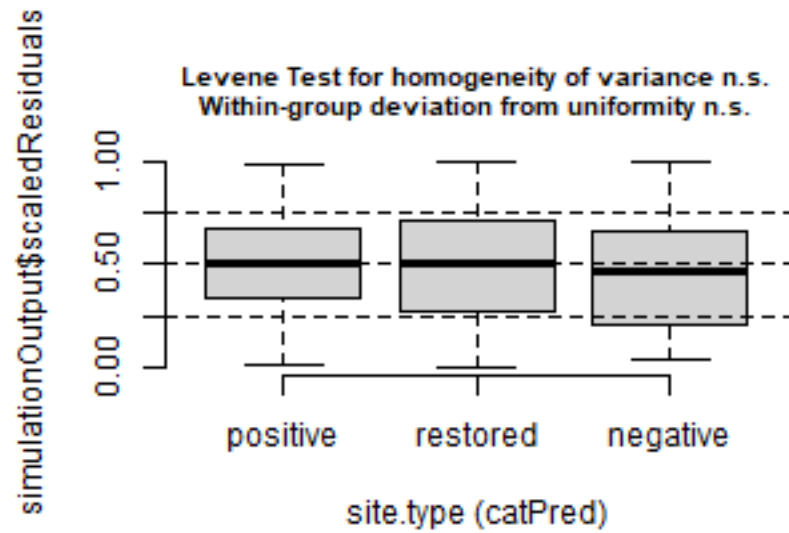
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



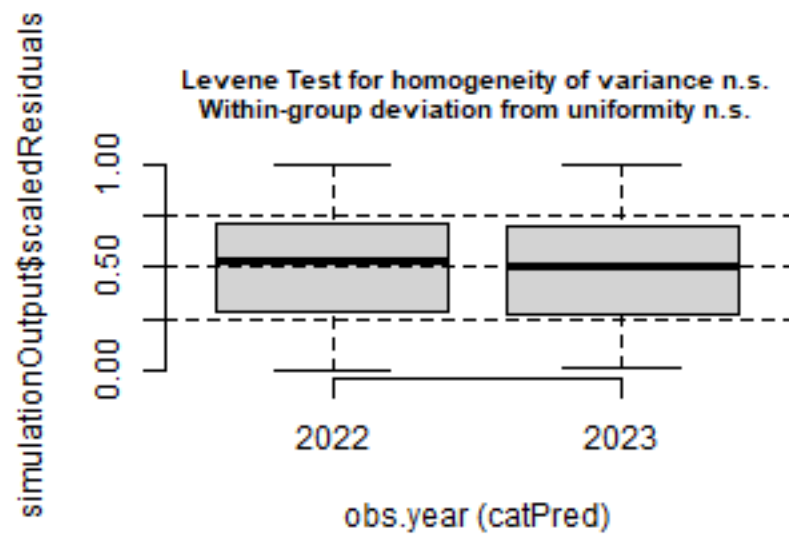
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



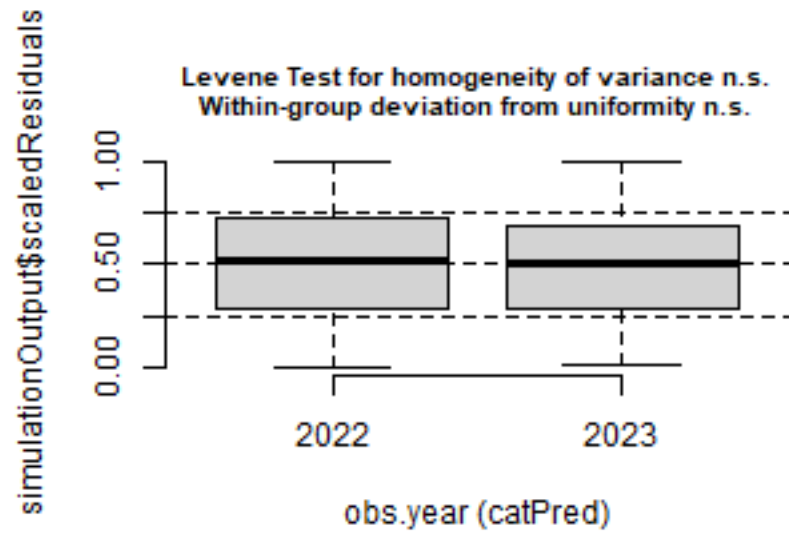
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



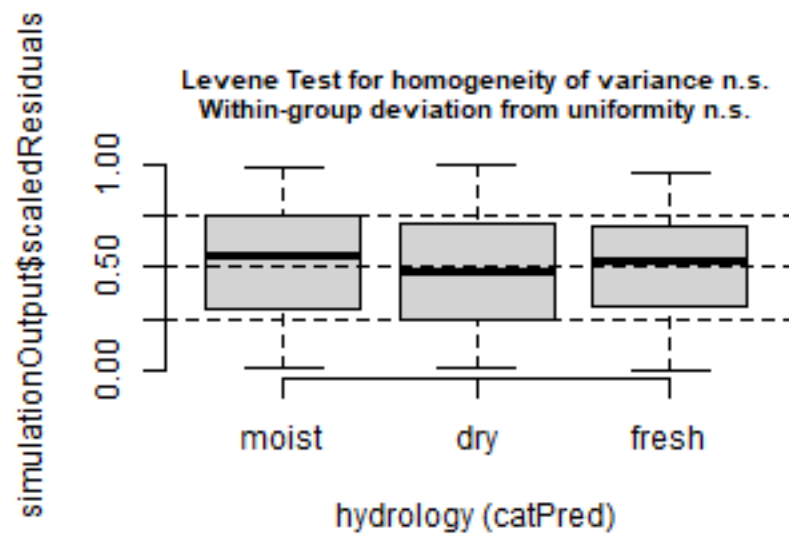
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



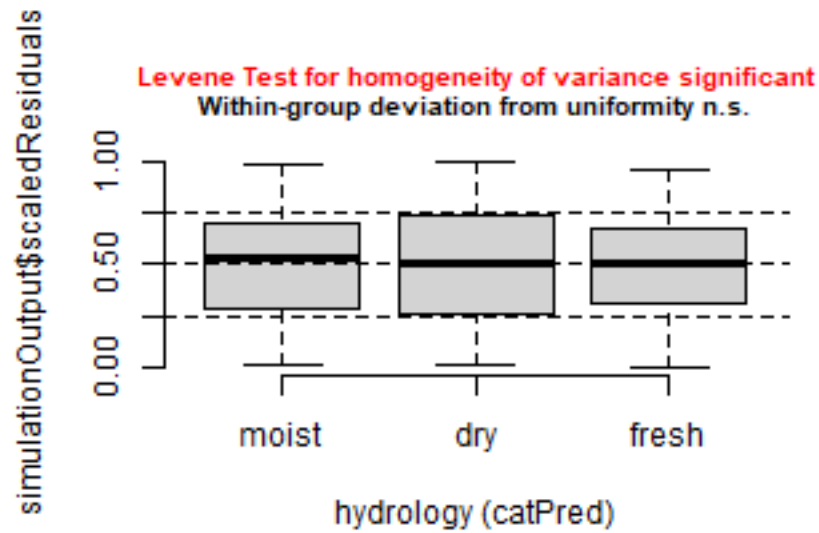
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



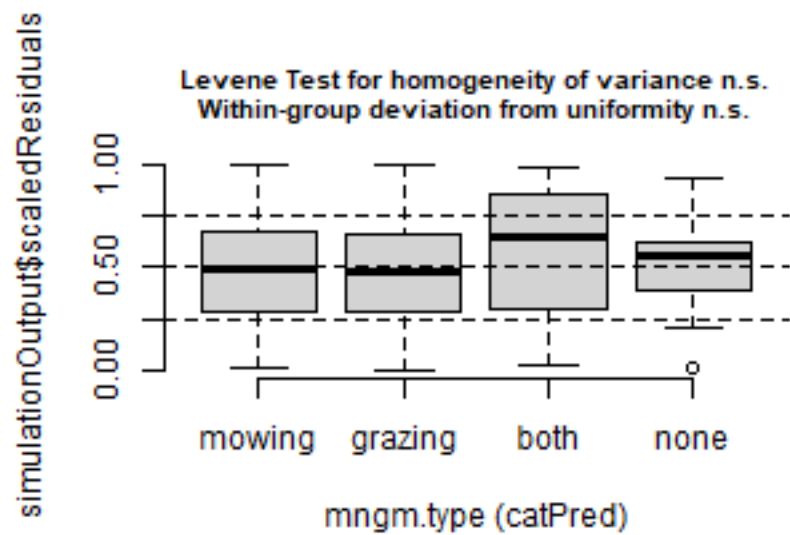
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



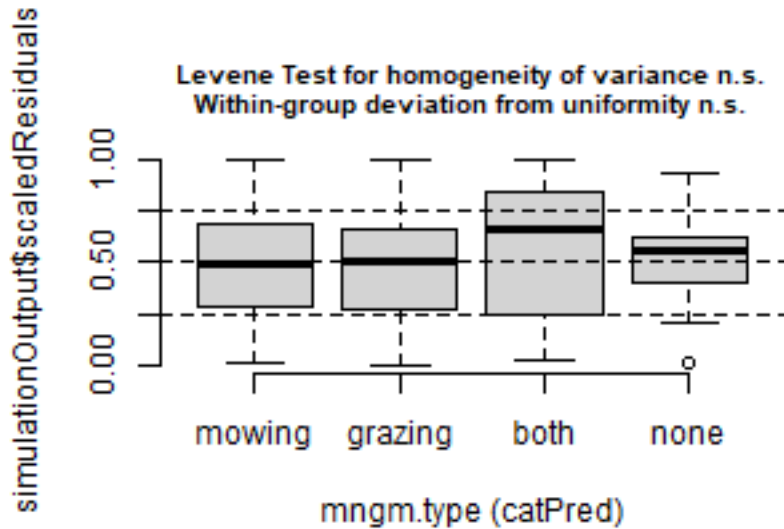
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
##   method      from
##   na.action.merMod lme4

##              GVIF Df GVIF^(1/(2*Df))
## esy4          5.565398 1      2.359110
## site.type     59.808391 2      2.780933
## eco.id        21.621706 2      2.156366
## obs.year      6.272316 1      2.504459
## esy4:site.type 9.503280 2      1.755773
## esy4:eco.id    2.591735 1      1.609887
## esy4:obs.year  2.989081 1      1.728896
## site.type:eco.id 80.706021 4      1.731264
## site.type:obs.year 9.596825 2      1.760078
## eco.id:obs.year 11.546974 2      1.843389
```

```
car::vif(m_2)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## esy4          70.853521 1      8.417453
## site.type     610.109338 2      4.969949
## eco.id        747.809303 2      5.229350
## obs.year      15.454263 1      3.931191
## hydrology     571.553093 2      4.889497
## esy4:site.type 90.941661 2      3.088095
## esy4:eco.id    22.256369 1      4.717666
## esy4:obs.year  5.791756 1      2.406607
## esy4:hydrology 11.653035 2      1.847608
## site.type:eco.id 186.649530 4      1.922553
## site.type:obs.year 15.336189 2      1.978925
## site.type:hydrology 778.808602 4      2.298417
```

```
## eco.id:obs.year      13.440637  2      1.914719
## eco.id:hydrology    2175.750423  4      2.613371
## obs.year:hydrology   29.116885  2      2.322931
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##           R2m      R2c
## [1,] 0.5488241 0.76601
MuMIn::r.squaredGLMM(m_2)
##           R2m      R2c
## [1,] 0.5910165 0.7728105
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnham & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_1 21 616.9389
## m_2 35 634.0819
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: sqrt(y)
##              Chisq Df Pr(>Chisq)
## (Intercept)    3297.8760  1 < 2.2e-16 ***
## esy4           13.0999  1 0.0002953 ***
## site.type       7.4542  2 0.0240625 *
## eco.id          17.6764  2 0.0001451 ***
## obs.year        2.1138  1 0.1459744
## esy4:site.type   3.6161  2 0.1639719
## esy4:eco.id       0.6128  1 0.4337345
## esy4:obs.year     2.1890  1 0.1389955
## site.type:eco.id  2.6650  4 0.6153506
## site.type:obs.year 11.7866  2 0.0027579 **
## eco.id:obs.year   13.8627  2 0.0009767 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_1)
```

```
## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: sqrt(y) ~ (esy4 + site.type + eco.id + obs.year)^2 + (1 | id.site)
## Data: sites
##
##      AIC      BIC    logLik -2*log(L)  df.resid
```



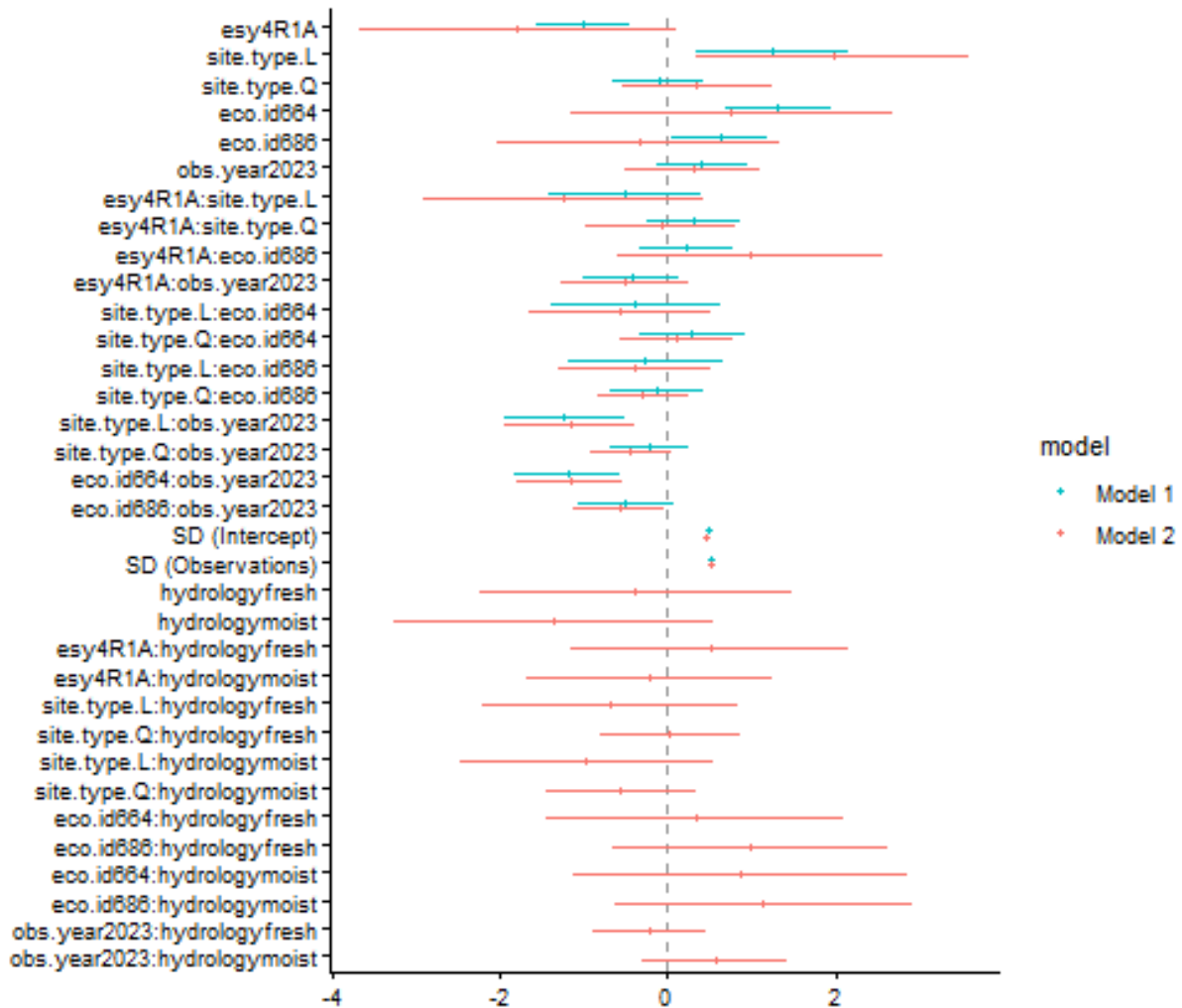
```
##      613.5      690.6     -285.8      571.5      270
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4341 -0.6138  0.0726  0.5302  3.0984
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
## id.site   (Intercept) 0.2437   0.4937
## Residual                0.2626   0.5124
## Number of obs: 291, groups: id.site, 119
##
## Fixed effects:
##                                Estimate Std. Error t value
## (Intercept)                   14.8624     0.2588  57.427
## esy4R1A                       -1.0223     0.2825  -3.619
## site.type.L                    1.2472     0.4569   2.730
## site.type.Q                   -0.1114     0.2791  -0.399
## eco.id664                     1.3218     0.3144   4.204
## eco.id686                     0.6192     0.2837   2.183
## obs.year2023                  0.4058     0.2791   1.454
## esy4R1A:site.type.L          -0.5098     0.4635  -1.100
## esy4R1A:site.type.Q          0.3151     0.2843   1.108
## esy4R1A:eco.id686            0.2258     0.2885   0.783
## esy4R1A:obs.year2023        -0.4341     0.2934  -1.480
## site.type.L:eco.id664        -0.3818     0.5089  -0.750
## site.type.Q:eco.id664        0.2935     0.3211   0.914
## site.type.L:eco.id686       -0.2658     0.4675  -0.569
## site.type.Q:eco.id686       -0.1183     0.2832  -0.418
## site.type.L:obs.year2023    -1.2313     0.3627  -3.395
## site.type.Q:obs.year2023    -0.2128     0.2346  -0.907
## eco.id664:obs.year2023     -1.1972     0.3218  -3.720
## eco.id686:obs.year2023     -0.5056     0.2883  -1.754
##
## Correlation matrix not shown by default, as p = 19 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
##
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```

```
## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
## Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_1,
  revpairwise ~ esy4,
  type = "response"
))
```

ESY EUNIS Habitat type

```

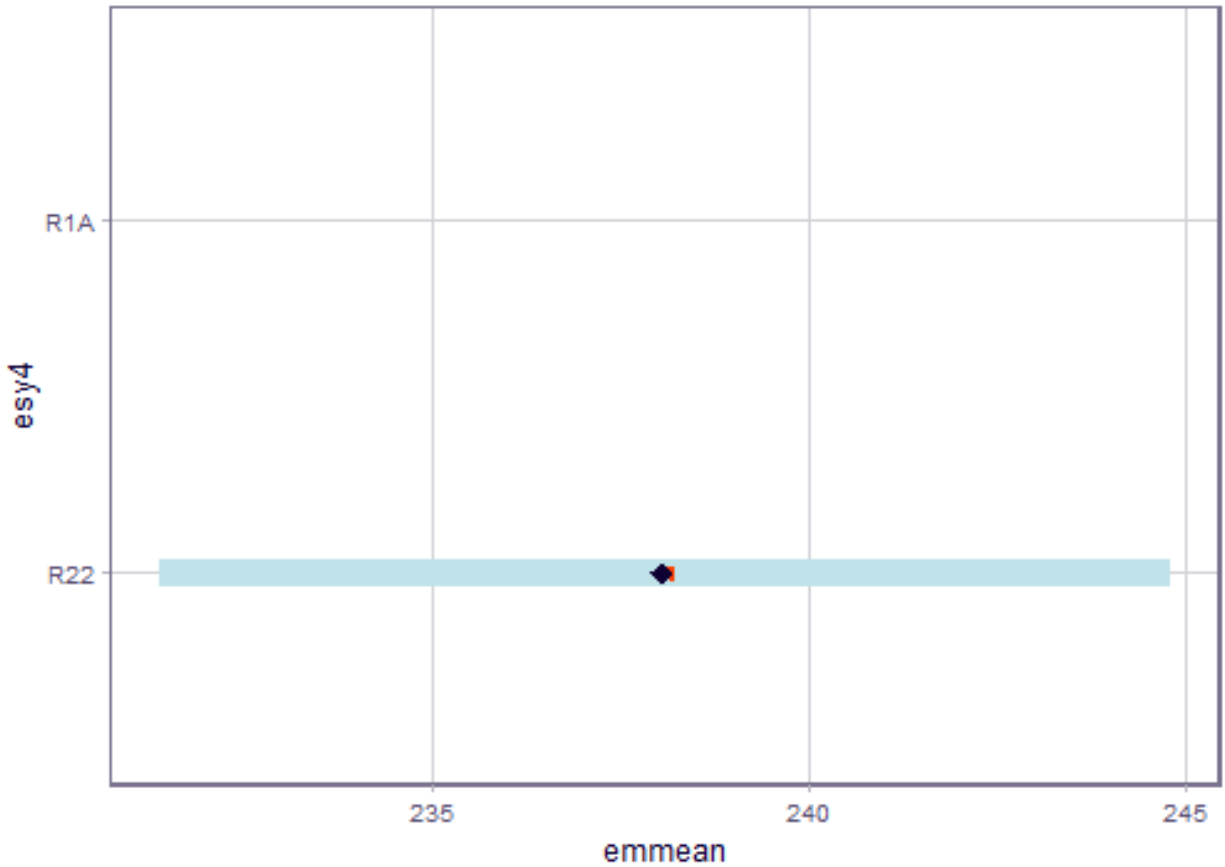
## $emmeans
##   esy4 response  SE  df lower.CL upper.CL
##   R22         238 3.4 199      231      245
##   R1A    nonEst  NA  NA       NA       NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the sqrt scale
##
## $contrasts
##   contrast estimate SE df z.ratio p.value
##   R1A - R22    nonEst NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Note: contrasts are still on the sqrt scale. Consider using
##       regrid() if you want contrasts of back-transformed estimates.
## Degrees-of-freedom method: kenward-roger
plot(emm, comparison = TRUE)

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

```



```
(emm <- emmeans(
  m_1,
  revpairwise ~ eco.id | esy4,
  type = "response"
))
```

Habitat type x Region

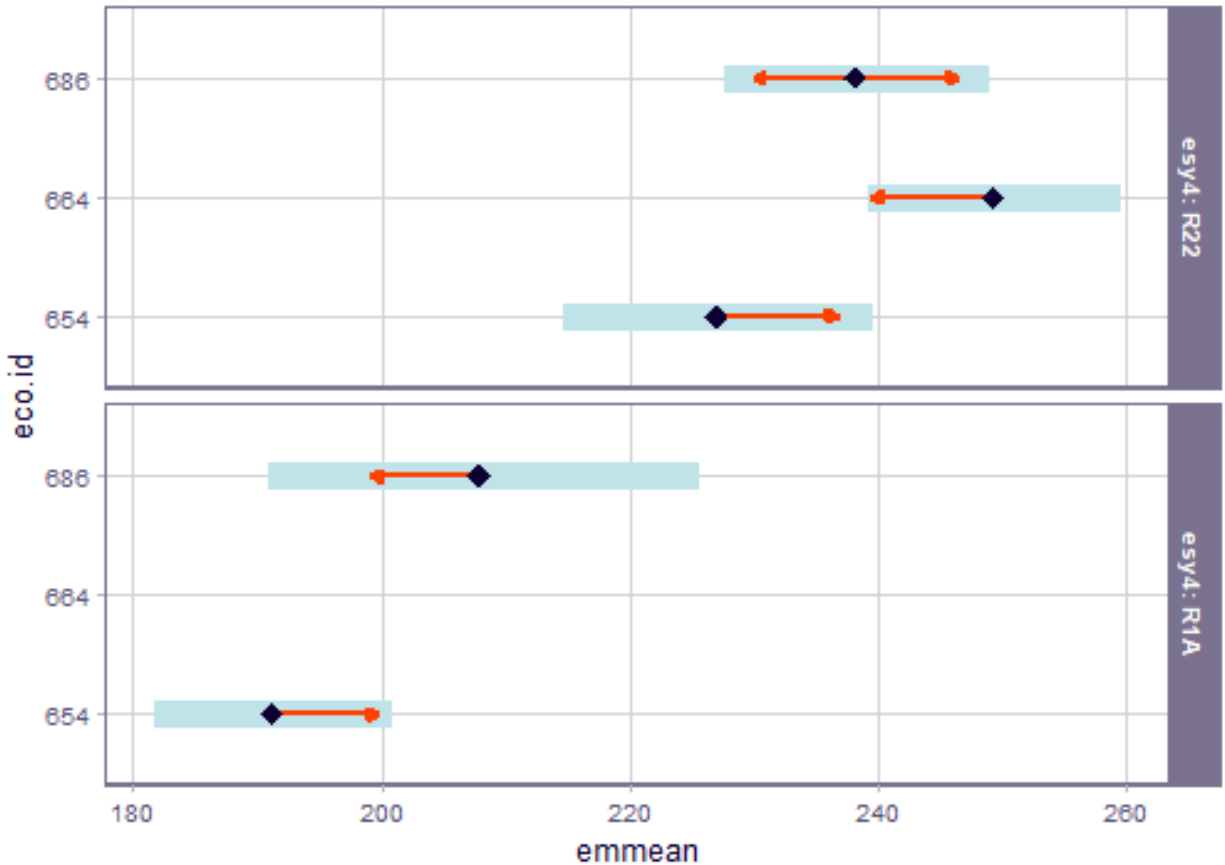
```
## $emmeans
## esy4 = R22:
##   eco.id response    SE  df lower.CL upper.CL
##   654         227 6.29 209      215      240
##   664         249 5.17 149      239      260
##   686         238 5.40 179      228      249
##
## esy4 = R1A:
##   eco.id response    SE  df lower.CL upper.CL
##   654         191 4.88 147      182      201
##   664      nonEst   NA   NA       NA       NA
##   686         208 8.78 186      191      226
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the sqrt scale
```

```
##
## $contrasts
## esy4 = R22:
## contrast estimate SE df t.ratio p.value
## eco.id664 - eco.id654 0.723 0.268 184 2.694 0.0210
## eco.id686 - eco.id654 0.366 0.249 176 1.474 0.3059
## eco.id686 - eco.id664 -0.357 0.244 163 -1.461 0.3126
##
## esy4 = R1A:
## contrast estimate SE df t.ratio p.value
## eco.id664 - eco.id654 nonEst NA NA NA NA
## eco.id686 - eco.id654 0.592 0.311 182 1.902 0.0587
## eco.id686 - eco.id664 nonEst NA NA NA NA
##
## Results are averaged over the levels of: site.type, obs.year
## Note: contrasts are still on the sqrt scale. Consider using
## regrid() if you want contrasts of back-transformed estimates.
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for varying family sizes
plot(emm, comparison = TRUE)

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] emmeans_2.0.3   DHARMa_0.5.0   patchwork_1.3.2 ggbeeswarm_0.7.3
## [5] lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0   dplyr_1.2.1
## [9] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2     tibble_3.3.1
```

```

## [13] ggplot2_4.0.3    tidyverse_2.0.0  here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.6      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.2.0       magrittr_2.0.5     multcomp_1.4-30
## [7] otel_0.2.0        compiler_4.5.2     mgcv_1.9-3
## [10] vctr_0.7.3        pkgconfig_2.0.3    crayon_1.5.3
## [13] fastmap_1.2.0     backports_1.5.1    labeling_0.4.3
## [16] utf8_1.2.6        ggstance_0.3.7     promises_1.5.0
## [19] rmarkdown_2.31    tzdb_0.5.0         nloptr_2.2.1
## [22] bit_4.6.0         xfun_0.58          later_1.4.8
## [25] broom_1.0.13      parallel_4.5.2     R6_2.6.1
## [28] gap.datasets_0.0.6 stringi_1.8.7       qgam_2.0.0
## [31] RColorBrewer_1.1-3 car_3.1-5           boot_1.3-32
## [34] estimability_1.5.1 Rcpp_1.1.1-1.1     iterators_1.0.14
## [37] knitr_1.51        zoo_1.8-15         parameters_0.29.1
## [40] httpuv_1.6.17     Matrix_1.7-4       splines_4.5.2
## [43] timechange_0.4.0  tidyselct_1.2.1    rstudioapi_0.18.0
## [46] abind_1.4-8       yaml_2.3.12        MuMIn_1.48.19
## [49] doParallel_1.0.17 codetools_0.2-20    lattice_0.22-7
## [52] plyr_1.8.9        bayestestR_0.18.1  shiny_1.13.0
## [55] withr_3.0.2       S7_0.2.2           evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3      pillar_1.11.1
## [61] gap_1.15.2        carData_3.0-6       foreach_1.5.2
## [64] stats4_4.5.2      insight_1.5.1       reformulas_0.4.4
## [67] generics_0.1.4    vroom_1.7.1         rprojroot_2.1.1
## [70] hms_1.1.4         scales_1.4.0        minqa_1.2.8
## [73] xtable_1.8-8      glue_1.8.1          tools_4.5.2
## [76] data.table_1.18.4 lme4_2.0-1          mvtnorm_1.4-0
## [79] grid_4.5.2        rbibutils_2.4.1     datawizard_1.3.1
## [82] nlme_3.1-168      Rmisc_1.5.1         performance_0.17.0
## [85] beeswarm_0.4.0    vipor_0.4.7         Formula_1.2-5
## [88] cli_3.6.6         gtable_0.3.6        digest_0.6.39
## [91] pbkrtest_0.5.5    TH.data_1.1-5       farver_2.1.2
## [94] htmltools_0.5.9   lifecycle_1.0.5     mime_0.13
## [97] bit64_4.8.2       dotwhisker_0.8.4    MASS_7.3-65

```